Res-Q Management System

Built as a course project for
CSE202: Fundamentals of Database Management Systems

Task 2: Design of Conceptual Model of App and converting it to Relational model

23rd January, 2026

Project Group No. - 18
Tutorial Group - 2
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❖ **Relational Schema and Er Model of our Model:**

❖ Link to Miro(Relational Schema): (last page)
https://miro.com/app/board/uXjVGMCke6g=/

❖ Link to Canva(ER Diagram). *NOTE : the attributes are mentioned in the docs*: (last page)
https://www.canva.com/design/DAG_OZNs1UM/l8W1C0FA2dXmlY_wqTLy8g/edit?utm_content=DAG_OZNs1UM&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

❖ **Relationships & Cardinality Constraints**

To effectively manage the Res-Q database, the following relationships are defined among the entities. Note that relationships marked with a **Green Circle (O)** in the diagram denote **Weak Relationships** involving weak entities.

- **Initiates** (User — Emergency Request)
- **Is Located In** (User — Polygon)
- **Contains** (Polygon — Micro-Hub)
- **Signs** (Worker — Contract)
- **Stations** (Worker — Micro-Hub)
- **Houses O** (Micro-Hub — Hub-Inventory) *[Weak Relationship]*
- **Is Listed In O** (Hub-Inventory — Part) *[Weak Relationship]*
- **Generates O** (Emergency Request — Dispatch Assignment)
- **Addresses O** (Dispatch Assignment — Worker) *[Weak Relationship]*
- **Billed Via O** (Emergency Request — Payment) *[Weak Relationship]*

❖ <u>Constraints</u>

A. Existential Constraints

- **Hub-Inventory** is a weak entity. It cannot exist without a physical **Micro-Hub** to store the items and a valid **Part Definition** to identify them.
- **Dispatch Assignment** cannot exist without a valid **Emergency Request** (to trigger it) and an active **Worker** (to fulfill it).
- **Payment** is strictly dependent on the existence of an **Emergency Request**; a bill cannot exist without a service request.
- **Worker Eligibility:** A Worker must have a signed **Contract** (verifying vehicle ownership) to be eligible for any assignment.
- **Spatial Constraint:** A **Micro-Hub** must be strictly located within a defined **Polygon** (Zone).

B. Cardinality Constraints (Min:Max)

- **User : Emergency Request (1:1 — 0:N):** A User can make zero or multiple emergency requests over time, but a specific request is initiated by exactly one User.
- **Worker : Contract (1:1 — 1:1):** A Worker signs exactly one active contract, and a specific contract belongs to exactly one Worker.
- **Worker : Micro-Hub (1:1 — 1:N):** A Worker is stationed at exactly one Hub per shift, whereas a single Micro-Hub serves as the base for many Workers.
- **Hub-Inventory : Part (1:1 — 0:N):** A specific inventory record (row) corresponds to exactly one Part Type. However, a single Part Type can be listed across multiple different Hub Inventories.
- **Emergency Request : Dispatch (0:1 — 1:1):** A Request generates exactly one Dispatch Assignment (if accepted). Pending or cancelled requests have zero dispatches.
- **Dispatch : Worker (1:1 — 0:N):** A Dispatch must be addressed by exactly one Worker. An experienced Worker can be assigned to multiple Dispatches over time.

❖ Relational Schemas

Underlined attributes represent Primary Keys (PK). Italicized attributes represent Foreign Keys (FK).

- **User** (UserID, Name, Email, Phone, Location_Coords, *Polygon_ID*)
- **Polygon** (PolygonID, ZoneName, BoundaryCoordinates)
- **Micro-Hub** (HubID, LocationAddress, *Polygon_ID*)
- **Worker** (WorkerID, Name, Contact_Number, VerificationStatus, AvailabilityStatus, *HubID*)
- **Contract** (ContractID, *WorkerID*, VehicleReg, VehicleModel, CleaningDutyClause)
- **Part** (PartID, Name, Category, Description, Price)
- **Hub_Inventory** (*HubID*, *PartID*, Quantity)
- **EmergencyRequest** (RequestID, *UserID*, Timestamp, IssueType, Status)

- **DispatchAssignment** (AssignmentID, *RequestID*, *WorkerID*, DispatchTime, Status)
- **Payment** (PaymentID, *RequestID*, Amount, TransactionStatus)

❖ Entity Attribute Specifications

This section details the specific attributes and data types for each entity as visualized in the schema diagram.

A. Core Entities

- **USER**
  - `UserID`: Integer **(PK)**
  - `Name`: Varchar(100)
  - `Phone`: Varchar(15)
  - `Email`: Varchar(100) **(Unique)**
  - `Location_Coords`: Varchar(100)
  - `Polygon_ID`: Integer **(FK)**
- **POLYGON**
  - `PolygonID`: Integer **(PK)**
  - `BoundaryCoordinates`: Varchar(255)
  - `ZoneName`: Varchar(100)
- **MICRO-HUB**
  - `HubID`: Integer **(PK)**
  - `LocationAddress`: Varchar(255)
  - `Polygon_ID`: Integer **(FK)**
- **WORKER**
  - `WorkerID`: Integer **(PK)**
  - `Name`: Varchar(100)
  - `Contact_Number`: Varchar(15) **(Unique)**
  - `VerificationStatus`: Varchar(50)
  - `AvailabilityStatus`: ENUM ('Standby', 'Busy', 'Off-Shift')
  - `HubID`: Integer **(FK)**

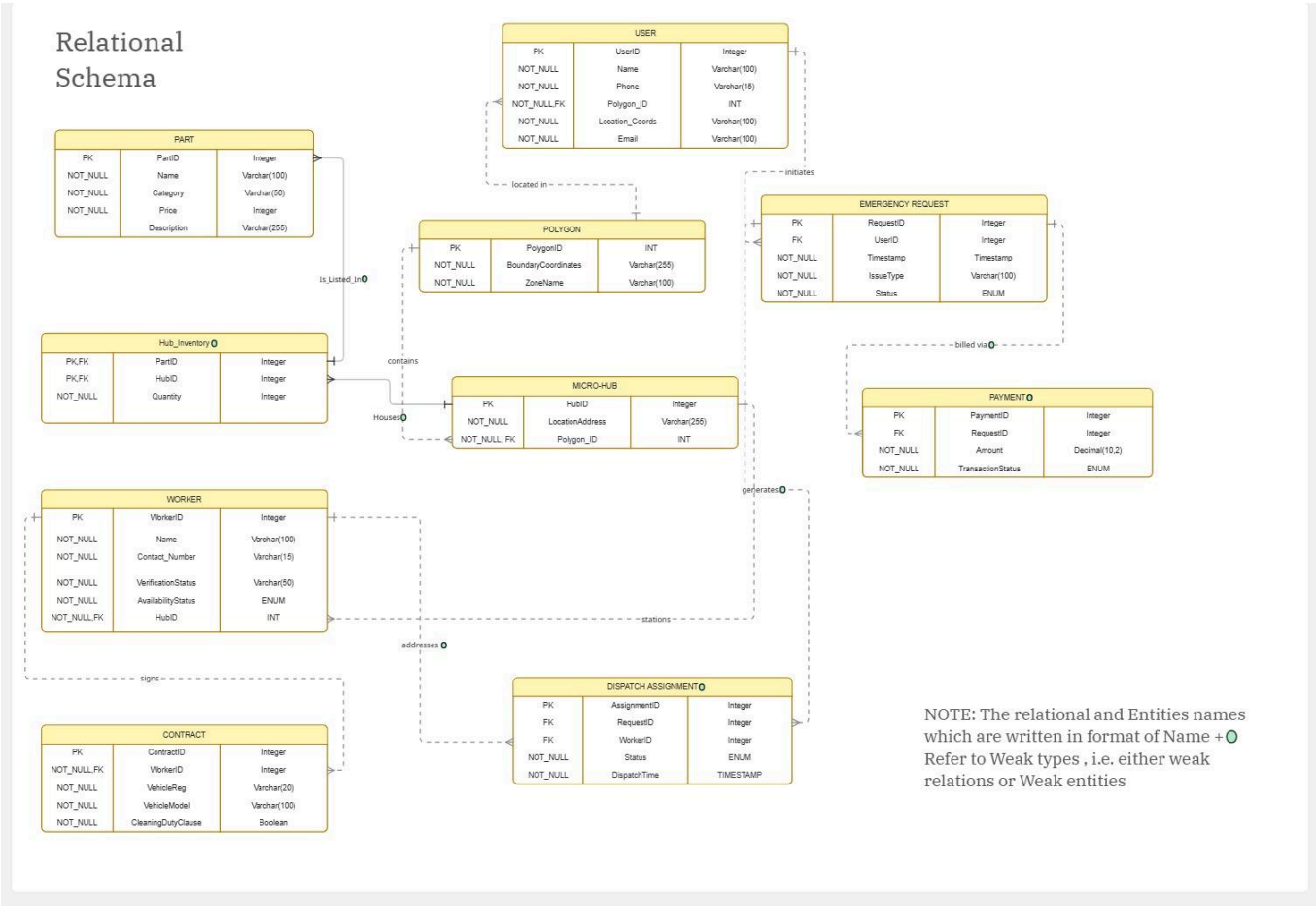
- **CONTRACT**
  - `ContractID`: Integer **(PK)**
  - `WorkerID`: Integer **(FK)**
  - `VehicleReg`: Varchar(20)
  - `VehicleModel`: Varchar(100)
  - `CleaningDutyClause`: Boolean
- **PART**
  - `PartID`: Integer **(PK)**
  - `Name`: Varchar(100)
  - `Category`: Varchar(50)
  - `Price`: Integer
  - `Description`: Varchar(255)

B. Weak Entities & Operational Tables

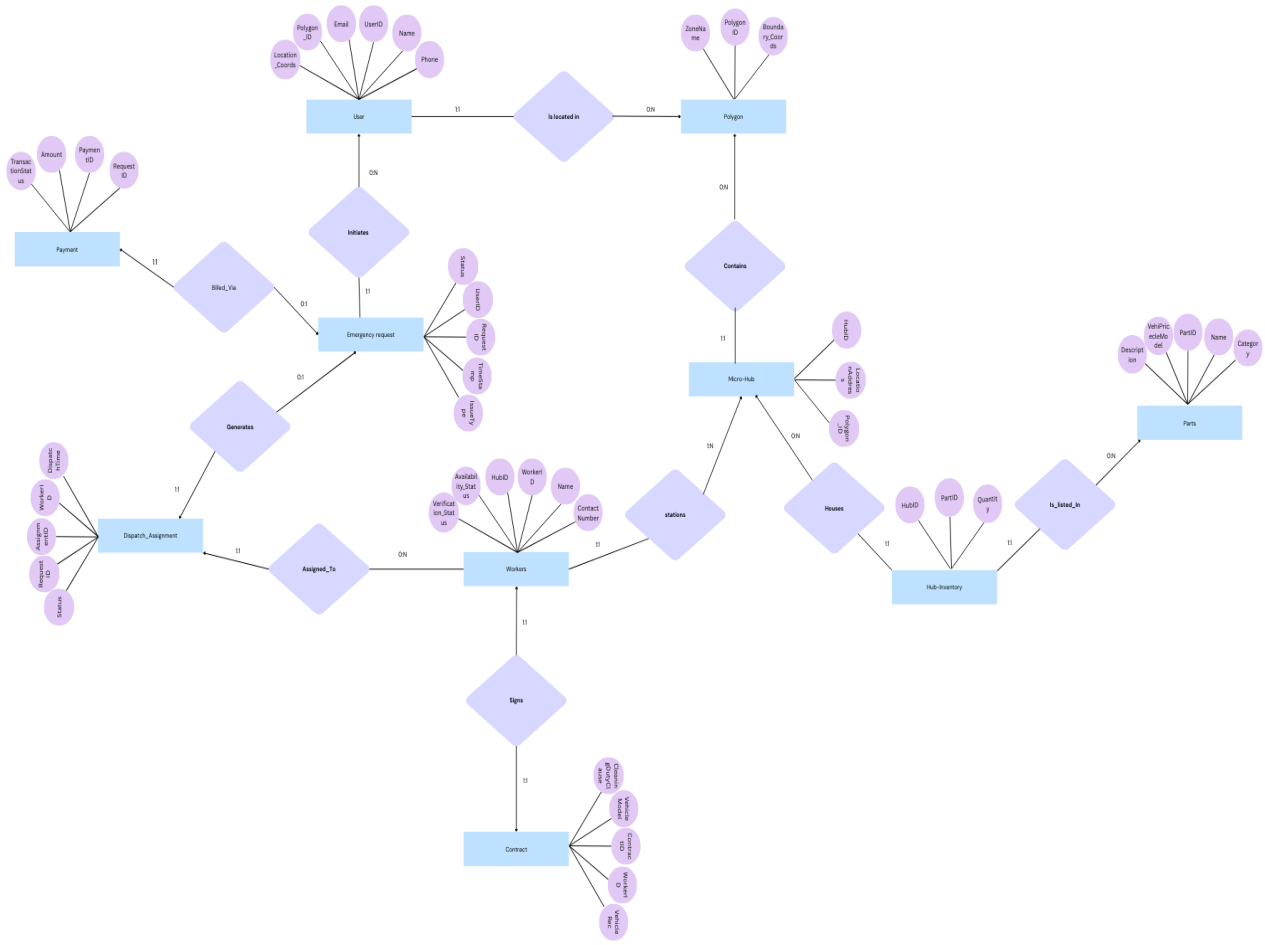
- **HUB_INVENTORY** (Weak)
  - `HubID`: Integer **(PK, FK)**
  - `PartID`: Integer **(PK, FK)**
  - `Quantity`: Integer
- **EMERGENCY REQUEST**
  - `RequestID`: Integer **(PK)**
  - `UserID`: Integer **(FK)**
  - `Timestamp`: TIMESTAMP
  - `IssueType`: Varchar(100)
  - `Status`: ENUM ('Pending', 'Completed', 'Cancelled')
- **DISPATCH ASSIGNMENT** (Weak)
  - `AssignmentID`: Integer **(PK)**
  - `RequestID`: Integer **(FK)**
  - `WorkerID`: Integer **(FK)**
  - `Status`: ENUM ('EnRoute', 'Arrived')
  - `DispatchTime`: TIMESTAMP
- **PAYMENT** (Weak)
  - `PaymentID`: Integer **(PK)**

- `RequestID`: Integer **(FK)**
- `Amount`: Decimal(10,2)
- `TransactionStatus`: ENUM

Relational Schema:



ER-Diagram


User
Polygon_ID
Email
UserID
Name
Phone
Location_Coords
1:1
Is located in
0:N
Polygon
ZoneName
Polygon ID
Boundary_Coords
0:N
Initiates
1:1
Contains
1:1
Payment
Transaction_Status
Amount
PaymentID
Request ID
1:1
Billed_Via
0:1
Emergency request
Status
UserID
Request ID
Timestamp
Urgency
0:1
Generates
1:1
Micro-Hub
HubID
Location_Address
Polygon ID
0:N
Parts
Description
VehPricedeModel
PartID
Name
Category
Dispatch_Assignment
Dispatch_Assignment_ID
WorkerID
Assignment_Timestamp
RequestID
Status
1:1
Assigned_To
0:N
Workers
Availability_Status
HubID
WorkerID
Name
Verification_Status
Contact Number
1:1
stations
1:N
Houses
HubID
PartID
Quantity
1:1
Hub-Inventory
1:1
Is_listed_in
0:N
1:1
Signs
1:1
Contract
Classification_Active
Vehicle Models
ContractID
WorkerID
Vehicle Info